AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION

System Component: Battery Management System (BMS) Architecture

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1. System Overview

Specifies the high-voltage 800V Traction Battery Management System software architecture. Enforces Cell Voltage Monitoring, State of Charge (SoC), State of Health (SoH), High Voltage Contactor Control, and Thermal Runaway Mitigation.

2. Software Components

Component Name (SWC)	SWC Type	Safety Level	Responsibility
`SWC_CellMonitor`	Sensor SWC	ASIL C	Samples 96 cell voltages and 24 NTC temperature sensors over isoSPI bus.
`SWC_StateEstimation`	Application SWC	ASIL C	Extended Kalman Filter for SoC, SoH, and Power Limits calculation.
`SWC_ContactorManager`	Actuator SWC	ASIL D	Main Positive, Main Negative, and Precharge contactor sequencing.
`SWC_ThermalRunawayGuard`	Safety SWC	ASIL D	Detects off-gas pressure spikes and triggers high-voltage pyrofuse.